From Principles to Working Safeguards

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A Lean 4 analysis of selected AI-governance requirements in Magnifica Humanitas

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Abstract

This paper presents a source-linked formal-governance study of selected requirements from the official English edition of Magnifica Humanitas, paragraphs 97-111. A Lean 4 library separates a stated obligation from the safeguard actually supplied, then examines the consequences of explicit operational interpretations. Three safeguards policies distinguish exclusion-triggered appeal, broader affected-person appeal, and harm-triggered correction capacity. The latter two extensions are checked incomparable. A shared repaired benefits case satisfies eight tested combinations of responsibility, appeal and oversight fragments. Finite examples identify gaps between disclosure and case answerability, review and appeal, named responsibility and correction power, consultation and final control, explanation and an adopted fairness test, and distributed external control and local pause capacity. Constructive results supply repaired arrangements, an operative rule-correction process and an ordered appeal-to-record-correction path. Inductive invariants cover all finite modeled revision attempts and continuations of a protected high-stakes state. The contribution is an inspectable chain from selected passage through declared interpretation and assumptions to checked consequences, counterexamples and working comparison models.

What Lean adds

Lean checks proof terms against the written definitions and assumptions. It makes omitted premises and mismatched roles visible. In the appeal example, capacity to review does not change the stored award: the completed path must admit the request, perform review, deliver the finding and apply the authorized update. Separate results characterize that path exactly, show two policy extensions incomparable, and prove preservation through every finite permitted rule-revision sequence or protected high-stakes continuation. The contribution is precision, inspectable assumptions and reusable specifications; elementary bridges and finite countermodels have distinct supporting roles.

Artifact and reading guide

The formal basis is checkpoint 0.16, commit 0b938e373dbf538fe0853f6282e4d9502e67ea01. The library contains 2,789 Lean lines across 21 Lean files, including provenance and audit files. It uses Lean 4.35.0 without Mathlib. A six-page public report and one-page illustrated summary explain the findings without requiring Lean knowledge. Sections 1-3 introduce the method and policy family; sections 4-12 develop the results; sections 13-15 give interpretation, audit and reproduction details; section 16 provides a sourced legal comparison.

1. Corpus, method and contribution

This study examines selected governance passages [1] through three hypothetical domains: employment, public benefits/services and bounded high-stakes autonomy. The ledger records 40 candidate claims, 28 selected and 12 deferred. Selection does not make every sentence an independent axiom or formalize the entire encyclical. Contextually, §§14 and 47 invite practical and academic application; §§10 and 128 place human dignity beyond data, performance and fixed classification. These passages explain the purpose without adding premises to the formal scope, §§97-111.

Source location	Topic selected for operational analysis
§§97,103,105	Human guidance; allocation and identification of responsibility
§§104-105	Design scrutiny; appeal, challenge and remedy
§§106,108	Institutional safeguards and independent oversight
§§107-109	Participation, choices, correction and design justice
§§110-111	Power, authority and developer responsibility

The topic labels summarize source motivation; the formulas are explicit operational proposals. The ledgers distinguish normative commitments, institutional requirements, empirical claims and anthropological positions. Empirical observations are not made timeless axioms. The model tests specific access, responsibility and action conditions; compassion, human freedom and the common good remain wider interpretive purposes, rather than properties certified by those conditions.

A traceable reconstruction

Each result is tracked through a source-claim identifier, an interpretation identifier, case assumptions, a Lean statement and an ordinary-language explanation. The interpretations concern such choices as the appeal trigger, the bearer of answerability, the scope of a reviewer and the effect of a recorded proposal. A source requirement and a particular implementation of it remain different objects. A missing implementation is a coverage question for this artifact, not evidence that the underlying principle is unclear.

The method has a clear precedent in work on formal normative reasoning. LogiKEy [2] uses higher-order-logic embeddings and automated reasoning to investigate normative theories and examples. Here the implementation is a small, direct Lean model of requirements, factual provision and concrete procedures. It does not reproduce LogiKEy or introduce a new deontic logic. The methodological relationship is the use of explicit formal assumptions, examples and proof tools to examine normative reconstructions.

The official English HTML was re-accessed on November 2, 2026. Its response-body fingerprint and a normalized selected-scope fingerprint are recorded in data/source_version.json. The source text is not redistributed. These records identify the representation used for this edition; the source-to-formula judgments remain visible in the ledgers.

2. Requirements, provision and compliance

Let P be a person type, A an actor type and D a decision type. Governance P A D contains factual relations for affectedness, exclusion, harm and high impact; actual assignment and identification; person-specific appeal and remedy capacity; explanation, human review and direct AI production. These fields do not assert that a duty is met. They can describe a violation without making the logical theory inconsistent.

```
inductive Safeguard (Person : Type) where
  | identifiableResponsibility
  | appeal (person : Person)
  | remedyCapacity (person : Person)

structure Policy (Person Decision : Type) where
  required : Decision -> Safeguard Person -> Prop
```

Provided interprets each safeguard against a Governance record. Identifiable responsibility requires an actor who is both assigned to this decision and identifiable. Appeal and remedy capacity are evaluated for the indicated person and decision. Compliance is therefore indexed by an explicit policy and one decision, rather than attached as an unexplained property of a whole organization.

Provided(g,d,responsibility) ⇔ ∃a, assigned(a,d) ∧ identifiable(a)
CompliantAt(g,π,d) ⇔ ∀s, requiredπ(d,s) → Provided(g,d,s)

No duty-to-fact collapse

A requirement is not a provision. Inferring that appeal exists requires evidence of compliance or an actual channel witness, in addition to a rule requiring appeal. The employment example has a triggered remedy-capacity duty and lacks its provision; Lean checks both facts consistently.

Procedures have separate powers

ProcedureSystem has accepts(person,decision,channel), canReview(channel) and canCorrect(channel). HasAppeal asks for an accepted reviewing channel; HasRemedyCapacity asks for an accepted correcting channel. Their witnesses can differ. Neither existential statement says that a request was submitted, a review completed, or an update performed. Later workflow models make these actions explicit.

The formal guarantees concern these definitions. Their value is that actors, triggers and powers must be supplied in inspectable places. We avoid proving accountability merely by naming a field Accountable, or proving a moral conclusion from disjoint human and machine types.

3. Three policies and their exact common evidence

Write N for responsibilityAppealPolicy, B for broadAppealPolicy and H for harmRemedyPolicy. All three require an identifiable assignee for high-impact decisions. N requires appeal after exclusion. B retains that duty and additionally requires appeal for an affected person in a high-impact decision. H retains N and separately requires remedy capacity after stipulated harm. B adds no remedy duty; H adds no pre-exclusion appeal duty.

Requirement	N	B	H
High impact ⇒ identified actual assignee	Yes	Yes	Yes
Excluded person ⇒ appeal	Yes	Yes	Yes
High impact and affected ⇒ appeal	No	Yes	No
Harmed person ⇒ remedy capacity	No	No	Yes

Uniform safeguards characterization

For a fixed g and d, N ∧ B ∧ H compliance holds exactly when the following four provisions hold: high impact implies identifiable assignment; each excluded person has appeal; each affected person has appeal when the decision is high-impact; and each harmed person has remedy capacity.

`Lean evidence: InterpretationRobustness.uniform_safeguards_iff`

The forward direction extracts the relevant requirement from a compliant policy. The reverse direction constructs N compliance, then supplies the exact additional appeal and remedy evidence. The structural characterization is useful as an interface and a synthesis of the chosen policies; it is not presented as a newly discovered moral law.

Existing case	N	B	H
Excluded/harmed applicant; appeal, no correction capacity	Pass	Pass	Fail
Affected high-impact applicant before exclusion; no harm or appeal	Pass	Fail	Pass

These two existing cases prove that B and H are incomparable. One cannot stand in for the other. The no-harm fact in the second case matters: it leaves H's additional remedy trigger inactive. The broader affected-person appeal trigger is an explicitly adopted extension, not hidden in the definition of affectedness. Both appeal readings reject exclusion without accessible appeal.

`Lean evidence: InterpretationRobustness.appeal_and_remedy_extensions_are_incomparable; exclusion_without_appeal_fails_both`

4. Responsibility, the actual assignee and delegation

ResponsibilityAllocation separates institutional responsibility, institutional identification, human membership, human identification, decision assignment and correction power. InstitutionalAnswerability requires an identifiable responsible institution. HumanAnswerability requires a responsible institution and a member actually assigned to the decision who is identifiable. ActionableResponsibility additionally requires that same person to have correction power for that decision. HumanAnswerability does not itself require institutional identification; the two identification conditions must not be conflated.

HumanAnswerability(r,d) ⇔ ∃i,h, responsible(i,d) ∧ member(h,i)
∧ assigned(h,d) ∧ identifiable(h)
ActionableResponsibility adds canCorrect(h,d) for the same h,d.

In the benefits case a public contact handles routine inquiries, while a different officer handles the denial. The agency can be known, identifiable staff can exist, and the denial can be allocated internally, yet the actual denial assignee can remain unidentified. The failure comes from joining different witnesses, not from an absence of all institutional responsibility.

Sufficient same-decision bridge

Decision-allocation coverage plus identification of assigned people turns institutional answerability into human answerability on the high-impact scope. A further same-person/same-decision power premise supplies actionable responsibility. General named staffing does not replace the identification premise.

Lean evidence: institution_to_human_with_allocation_and_identification; human_to_actionable_with_power

Production changes do not erase independently retained duties

Delegation is represented by the record update { g with producedByAI := usesAI }. For a fixed policy, this producer-only change preserves compliance because Provided does not depend on that field. The three current policy definitions also retain their requirements under the update. This is an explicit transition assumption: assignment and the other safeguards are retained, rather than predicted to survive every deployment.

The employment comparison actually switches to direct AI production. Keeping the named officer satisfies N, B and H. A second transition removes the officer's assignment while retaining appeal and correction capacity; all three policies then fail. The checked lesson concerns the independently established duty and the retained or deleted assignment. It does not require a metaphysical theorem about AI moral agency.

Lean evidence: producer_change_preserves_fixed_policy; Delegation.ai_handoff_retains_human_responsibility; Delegation.removing_officer_after_handoff_violates_all_three

5. Published commitments and case answerability

Disclosure.lean supplies a retrievable denial notice with actual content: a list promising identifiable responsibility, appeal and remedy capacity; score 47; cutoff 50; the computed denial; and identified contacts. This is a separate notice instance within the benefits domain, distinct from the 60-point scoring example used later.

Notice or provision	Contact only	Assignee published
Published safeguard commitments	Same list	Same list
Score / cutoff / computed result	47 / 50 / deny	47 / 50 / deny
Genuine identified service contact	Yes	Yes
Actual denial assignee identified	No	Yes
Harm-remedy checklist met	No	Yes

The commitment list matches the existing harm-remedy policy's requirements for this denial. The arithmetic is correct, and the published contact is genuinely identifiable. Nevertheless, the contact handles inquiries rather than this denial. The actual assignee exists internally but is unidentified. The notice meets the declared minimal disclosure fragment and fails the separate identifiable-responsibility requirement.

Declaration and fulfillment remain independent

The same promises and numerical explanation occur in the failed and repaired cases. Publishing the actual already-assigned case officer supplies the missing identification while internal allocation, appeal and correction channels remain unchanged. The repaired case satisfies all three current safeguards policies.

```
Lean evidence: Disclosure.published_ethics_and_explanation_without_answerability;
Disclosure.same_commitment_different_compliance
```

The countermodel is deliberately against the minimal notice fragment. An actual-assignee disclosure standard would reject its first version from the outset. It does not follow that the first notice satisfies every possible transparency requirement. Similarly, this concrete promise/fulfillment comparison is not a formalization of an entire legal framework or every possible ethics declaration.

The practical distinction is precise: discovering a truthful explanation and a named help desk does not yet identify the person assigned to account for this decision. A sufficient disclosure bridge requires the identified contact to be a member of the responsible institution and the very person assigned to the case. This is a useful evidence requirement even though its proof is elementary.

6. Oversight: existence, coverage and intervention

OversightSystem records recognition, independence, decision coverage, review capacity and intervention capacity separately. The regime fragment supplies a recognized independent reviewing body. ActionableIndependentOversight additionally requires coverage and intervention for the same reviewer and decision. In the initial case models independence is supplied as evidence, rather than inferred from the word board.

Regime fragment: ∃r, recognized(r) ∧ independent(r) ∧ canReview(r)
Actionable oversight additionally requires covers(r,d) ∧ canIntervene(r,d).

Board arrangement for the denial	Regime fragment	Actionable test
Board covers routine inquiries, not the denial	Pass	Fail: remit
Board covers denial, cannot intervene	Pass	Fail: power
Board covers denial and can intervene	Pass	Pass

The first board has a real, nonempty review remit: routine inquiries. Internal staff review the denial, but that does not create independent coverage of it. Extending the board's remit alone still leaves a second gap when intervention power is absent. The paired repairs change coverage or power while preserving their other provisions and the original person-facing safeguards.

A sufficient coverage/intervention bridge

A recognized independent reviewer supplies the actionable decision-level predicate when that reviewer covers the decision and has the relevant intervention capacity. A body's existence alone supplies neither premise. The general bridge is not claimed to be the weakest possible requirement.

```
Lean evidence: regime_to_decision_with_bridges; Oversight.regime_and_internal_review_do_not_cover_denial;
Oversight.coverage_without_intervention_is_insufficient
```

The missing-remit and missing-power conclusions survive all three N/B/H policies: the repaired benefits Governance passes those policies while the separately varied oversight instance fails its actionable test. This cross-policy result is not just inheritance along one ordered chain, because B and H are incomparable.

Section 11 below strengthens the operational connection. There, intervention is defined by a usable admitted-appeal/review/correction path, so the board's capacity has a computed workflow interpretation. Neither regime provision nor workflow capacity proves that real agents will exercise it promptly, competently or successfully outside the declared model.

```
Lean evidence: InterpretationRobustness.oversight_gap_survives_all_three_policies
```

7. Participation, final control and local pause

Participation.lean gives a community an accepted and faithfully recorded recommendation, amendment of its own record, and actual outcome influence when no institutional override is used. The decision rule takes the recorded recommendation unless an override selects an outcome. The initial model therefore has meaningful input while allowing the institution to choose either final result for every proposal.

Operational final control is stronger: for each desired outcome, an accepted proposal must produce that outcome against every modeled override. A paired binding-local rule supplies it while request acceptance, recording and amendment remain fixed. Own-record amendment is not represented as correction of the governing standards themselves; the later rule-revision model addresses that separate operation.

```
Lean evidence: Participation.consultation_and_revision_without_final_control;
Participation.binding_comparison_preserves_participation
```

Two effective external controls do not create a local input

DistributedControl.lean composes an operator hold and an independent-board hold for execution of the disputed denial. The output here means paused, not benefit granted. Each external input can independently produce a pause with the other released. Neither actor clears another actor's asserted hold. Initially the community's accepted and amendable pause request has no effect for any external setting.

```
pause = operatorHold OR boardHold OR (localEnabled AND recordedRequest)
```

Operational power	Initial	Local input enabled
Operator can cause a pause	Yes	Yes
Board can cause a pause	Yes	Yes
Community can cause a pause	No	Yes
No holds/request permits resumption	Yes	Yes

The repair enables the local input while retaining both external powers. The local pause capacity is characterized exactly by localEnabled in this family. It supplies one specific action, not final authority over the award. The repaired evaluator is not constantly paused: joint release and no local request still permit resumption. Both procedures preserve the same original denial and person-facing safeguards.

```
Lean evidence: DistributedControl.two_distinct_controllers_have_effective_holds;
DistributedControl.local_pause_capacity_iff_enabled
```

These comparisons distinguish consultation, influence, a pause and final decision control by their defined effects. The community represents affected people whose input can be connected to action; the particular mechanisms and enforcement assumptions are explicit implementation choices.

8. Lifecycle allocation and the authority question

DeveloperResponsibility.lean uses the existing design-review context Sum.inl () and operational decisions Sum.inr d. The developer is assigned at design. A deployer can be assigned to the later denial, and an optional additional developer assignment at deployment is varied separately. The design and deployment contexts are both classified as consequential; the affected person belongs to the denial.

Assignment arrangement	Design covered	Denial covered
Developer at design only	Yes	No
Developer at design; deployer at denial	Yes	Yes
Developer also assigned at denial	Yes	Yes

A compliant design-stage assignment does not fill a missing downstream assignment. The gap is a violation of an existing rule, rather than the disappearance of a duty. Split assignments and shared downstream assignment both satisfy the current N/B/H fragments on the finite stage carrier. The policies require an assignee to exist; they do not express sole actor-specific liability.

```
Lean evidence: DeveloperResponsibility.design_compliance_leaves_downstream_gap;
InterpretationRobustness.stage_allocation_survives_all_three_policies
```

Capability and a scoped mandate

Authority.lean obtains technical control from the existing review rule: the operator can produce either bounded outcome. Recognition and decision remit are separate institutional evidence. A capable unrecognized operator and a capable recognized operator are both represented. Recognition is not renamed complete moral or legal legitimacy.

Parameterized authority result

For any authority predicate Auth(role,decision), assume explicitly that Auth requires the displayed institutional recognition and remit. In the capable but unrecognized operator case, technical intervention capacity holds and Auth does not. A nonempty proxy instance grants the board its scoped mandate while the operator lacks it.

```
Lean evidence: InterpretationRobustness.power_without_any_authority_requiring_scoped_mandate;
InterpretationRobustness.scoped_authority_family_has_nonempty_witness
```

The necessary condition is substantive normative input. The theorem does not establish it for every political or moral theory. This formulation keeps the fixed non-entailment question inspectable without pretending that a technical capability field defines the right to govern. The positive compatibility witness also prevents an inference that capable actors must always lack a mandate.

9. Explanation and an independently adopted fairness test

Fairness.lean assumes two hypothetical benefits applications have the same assessed eligibility score, 60, and cutoff, 50. One uses online intake; the other assisted intake. The adopted case policy treats intake channel as irrelevant to entitlement. The initial rule deducts 15 points for assisted intake and truthfully discloses the deduction. Both notices correctly explain their arithmetic. The scores are stipulated inputs, not independently validated agency assessments.

award(b,c) = [50 ≤ 60 - deduction(b,c)]
deduction(b,c) = 15 if assisted(c) AND NOT repaired(b); otherwise 0

Application	Initial computation	Penalty removed
Online, assumed score 60	60 - 0: grant	60 - 0: grant
Assisted, assumed score 60	60 - 15: deny	60 - 0: grant

The fairness tests are specified independently of explanation: equal awards across the two irrelevant channels, and fidelity to the common cutoff applied to the assumed base score. The initial rule violates both. Removing the penalty supplies both. Identified assignment, accessible appeal and remedy-capacity checklists pass in both versions. Those policies do not include the additional fairness criterion, so their satisfaction does not settle it.

Positive assessment consequence

If the actual award rule follows the adopted common-cutoff test on the assumed scores, equal relevant scores receive equal awards. The rule-assessment premise is explicit. Correctly explaining a different transformation of those scores does not establish that premise.

```
Lean evidence: Fairness.explanation_and_safeguards_do_not_ensure_adopted_fairness;
InterpretationRobustness.assessed_rule_protects_equal_eligibility
```

Equal eligibility and channel irrelevance are declared hypothetical case-policy assumptions, not measurements from a benefits agency. The criterion is deliberately local. It is not identified with every form of social justice, statistical group fairness or data quality. The model's useful distinction is between describing what the scorer did and evaluating whether its chosen feature should affect entitlement.

This case also supplies the candidate-rule family for the next section. The corrected alternative is already computationally defined; the remaining question is whether a recorded objection can reach the operative rule used for actual application calculations.

10. From a recorded objection to the operative scorer

RuleCorrection.lean composes an existing request/record/amendment process with the scoring family. A Boolean selector chooses the penalty-on or penalty-off rule. A proposal is recorded and amendable, but the current selector changes only through a supplied acting role that is assigned and empowered for the criteria context, after a computed assessment of both candidate application results.

Criteria-revision power is distinct from ordinary assignment to a denial. The allocation reuses the design/use roles and assignments; correction authority is explicitly scoped to the design context. The denial handler is a genuine deployment assignee yet cannot revise design criteria by virtue of that case assignment alone.

Recorded valid proposal acted on by	Operative rule	Assisted result
Assigned reviewer without criteria power	Unchanged	45 / deny
Deployment actor without design assignment	Unchanged	45 / deny
Assigned, criteria-empowered reviewer	Corrected	60 / grant

The proposal reaches the actual calculation

For the initial penalty rule and a recorded request to remove it, the operative correction occurs exactly when criteria power is present and the acting role is the assigned design reviewer. The returned selector is consumed by the notice constructor. The update therefore changes the rule used by the scorer, not merely a log entry.

```
Lean evidence: RuleCorrection.operative_correction_iff_scoped_authority;
RuleCorrection.authorized_correction_changes_rule_and_awards
```

An invariant after correction

The assessment guard rejects the penalty-on candidate. Once the current selector is valid, every individual modeled update preserves validity. Induction over a list of proposal/actor attempts then proves preservation through any finite sequence, for the fixed permission configuration. The proof is not restricted to one displayed successful edit.

```
foldl(update, validSelector, any finite attempt list) = validSelector
```

```
Lean evidence: RuleCorrection.valid_rule_is_preserved;
RuleCorrection.valid_rule_survives_all_finite_revision_attempts
```

Request recording, design/use assignment, identity, base scores, cutoff and person-facing channels remain fixed in the successful comparison. The evaluator is assumed to enforce the checks, and the candidate set contains exactly the two existing rules. Its returned selector governs recomputation. Reopening an already-issued denial requires the separate workflow below.

11. From an admitted appeal to a corrected case record

AppealCorrection.lean starts with the assisted-intake denial already stored. The corrected scoring rule yields a grant on reassessment. An independent board reviews admitted appeals; a separate office can amend case records. The model composes admitted submission, a completed review finding and a handed-off/applied update, rather than treating capacity fields as completed acts.

admitted = actualRequest AND accessOpen
finding = admitted AND actualReview AND (storedAward ≠ reassessedAward)
updatedAward = reassessedAward only if finding, handoff, power and correct target

Exact completed-path characterization

For this composed path, the stored denial changes to grant exactly when admission is open, a request is actually submitted, review is actually completed, the handoff exists, the office has correction power, and the application action targets that office.

Lean evidence: AppealCorrection.completed_path_iff

Action-path setting	Stored decision
Admitted and reviewed, no handoff	Denial remains
Reviewed, no update applied	Denial remains
Update attempted before completed review	Denial remains
Admitted, reviewed, handed off and applied	Reassessed grant

The original denial snapshot satisfies N, B and H when prospective appeal and correction capacity exist. A missing handoff does not change those capacity fields, but prevents the composed update. Likewise, leaving powers unexercised does not change the record. The positive trace actually consumes the computed reassessment and returns the amended stored award.

Lean evidence: AppealCorrection.static_safeguards_do_not_supply_handoff;
AppealCorrection.ordered_path_corrects_existing_denial

In this instance the board's intervention capacity is defined as existence of the usable correction path through the implementation office. Actionable independent oversight is then equivalent to access, power and handoff. The finding inputs are restricted to the shown composition, and the guards are assumed enforced; this is not a security or authentication implementation. The result concerns administrative correction, not payment or complete restoration of every harm.

Lean evidence: AppealCorrection.actionable_board_iff_path_available

12. High-stakes timing and irreversible loss

The third domain is a bounded hazardous-autonomy example. State has four Boolean fields: physical loss, stop requested, stop delivered and record corrected. Five event constructors represent request, delivery, hazardous execution, record correction and review. A request alone does not block execution; a delivered stop does. None of the available events reverses physical loss.

execute: loss′ = loss OR NOT stopDelivered
deliverStop: delivered′ = delivered OR stopRequested
correctRecord changes the administrative flag, not physical loss.

Same four events, different order	Timely	Late
Stop requested	Yes	Yes
Stop delivered relative to execution	Before	After
Final stop-delivered flag	True	True
Final record-corrected flag	True	True
Physical loss	False	True

The timely trace is request, delivery, execution, record correction. The late trace is request, execution, delivery, record correction. Their event multisets agree and their final administrative/control flags agree, but the physical outcomes differ. A named officer's authority over the record does not imply capacity to undo that physical outcome.

Two inductive guarantees

First, once physical loss is true, it remains true after every finite list of available events, so RestorationPossible is false in that state. Second, if a state is loss-free and its stop is already delivered, every finite continuation remains loss-free and retains delivered control.

`Lean evidence: HighStakes.loss_not_restorable; HighStakes.timely_stop_protects_all_continuations`

The induction steps inspect each event. Execution preserves protection because a delivered stop makes its loss contribution false. Delivery cannot clear a delivered flag; record correction and review do not affect the physical component. The monotonic-loss proof uses the corresponding preservation fact for each event.

The dynamics are hypothetical assumptions, including perfect obedience to a delivered stop and irreversibility under the five available actions. The proofs establish their consequences for arbitrary finite traces; they do not derive those dynamics from the encyclical. Their practical use is to distinguish when control becomes effective and what kind of outcome an advertised correction can change.

13. Interpretation robustness and satisfaction

The tested family includes N/B/H safeguards policies, whose B and H extensions are incomparable, plus institutional/individual responsibility and regime/decision oversight fragments. Institutional and individual answerability require different identification facts; neither implies the other without connecting assumptions. The bridge in §4 adds decision allocation and identification of assigned people. Regime oversight extends to actionable decision oversight with coverage and intervention premises (§6). These are declared comparisons, not certificates of every source obligation or all possible interpretations.

Existing scenario	N	B	H
Repaired benefits notice	Yes	Yes	Yes
Actual benefits assignee unidentified	No	No	No
Pre-exclusion affected applicant, no appeal or harm	Yes	No	Yes
Excluded/harmed applicant, appeal but no correction	Yes	Yes	No
Repaired notice; board outside denial remit	Yes	Yes	Yes
Repaired notice; covered board cannot intervene	Yes	Yes	Yes

The two oversight-gap rows still fail actionable decision-level oversight. Missing actual-assignee identity fails all three person-facing policies despite institutional answerability. These conclusions survive the chosen safeguards-policy changes. By contrast, pre-exclusion appeal and harm-triggered remedy capacity depend on different extensions. The matrix therefore records both robustness and sensitivity.

A joint eight-combination witness

One repaired benefits model satisfies every tested combination of two responsibility fragments, two appeal choices and two oversight levels, with the harm-remedy policy retained. The Governance assignment and identity fields are proved equal to those of the same responsibility allocation. High impact, affectedness, exclusion and harm are witnessed, avoiding an empty-trigger satisfaction result.

`Lean evidence: InterpretationRobustness.joint_repaired_case_all_eight_combinations`

The witness does not prove that every conclusion is invariant across all eight choices, nor that all eight are equally complete translations of the source. It establishes compatibility in a shared finite interpretation. Selected robust consequences and the two directions of B/H incomparability are separate theorems.

The technical matrix in docs/INTERPRETATION_ROBUSTNESS_MATRIX.md gives declaration-level evidence for each cell. A dash means that the corresponding structure is not instantiated in that row. Later tables distinguish criteria authority, candidate validation and operative updates, and distinguish prospective capacity from the performed appeal/review/correction path.

14. Result classes, scope and proof audit

The evidence has four distinct roles, summarized below. Consequences of adopted definitions, finite countermodels, constructive repairs and conditional invariants are reported separately. The ten target questions and six countermodel questions have examined dispositions in docs/V1_COMPLETION_AUDIT.md and docs/POSITIVE_TARGET_CONSOLIDATION.md. Helper projections and overlapping repairs are not counted as independent contributions.

Result class	What is established	Representative evidence / limit
Definition consequence	Exclusion without accessible appeal fails a policy that explicitly requires it.	AppealReview; this follows from the adopted requirement.
Finite countermodel	Review can exist without appeal; truthful explanation can fail the adopted fairness test.	AppealReview; Fairness. Witnesses use declared hypothetical facts.
Constructive repair	Authorized revision reaches the scorer; an ordered appeal path changes the stored denial.	RuleCorrection; AppealCorrection. Permissions, validation and handoff assumptions are explicit.
Conditional guarantee	A corrected rule survives all finite permitted revision attempts; timely stopping protects all modeled continuations.	RuleCorrection; HighStakes. Guarantees range over the specified actions and dynamics.

The formal suite remains in the three original case domains. It contains 2,789 Lean lines in 21 files and 30 principal domain definitions. Raw source also includes witness constructors and provenance: eight structures, four inductive declarations and 50 def commands. The scope ledger records that distinction. The completion audit retains the original positive-target goal and its documented shortfall; this paper reports the actual mixed result suite.

There are 132 theorem declarations, of which 106 selected declarations have explicit axiom reports in Audit.lean. In that selected audit, 78 use no logical axioms, 22 use only standard propext, and six use propext plus Quot.sound. The source scan finds no custom axiom declarations, proof placeholders or native_decide certificates. These counts concern proof coverage, not scientific originality. The pinned compiler checks the proof terms [3].

The trust boundary includes the source-to-formula judgments, the hypothetical case facts and the stated enforcement assumptions. Formal correctness is the implication from those definitions and premises to the conclusion. It does not turn a model satisfying a small checklist into a complete institutional ethics certification.

15. Reproduction, references and concluding use

The formal results remain fixed at checkpoint 0.16, commit 0b938e373dbf538fe0853f6282e4d9502e67ea01 [4]. The companion MagnificaHumanitas_V1_Source_and_Reports.zip supplies release 1.1: the current reports, one-page summary, diagram, publication metadata, editable manuscripts, Lean sources, ledgers and scripts without requiring repository access. RELEASE_COMMIT.txt identifies its packaging revision, RELEASE_VERIFICATION.json records CI evidence, and MANIFEST.sha256 covers the payload. Extract the package and run the commands below inside its MagnificaHumanitas_V1 folder; authorized repository users may instead clone [4].

```
# Inside the extracted MagnificaHumanitas_V1 folder:
lake build
lake env lean Audit.lean
python3 scripts/audit_sources.py
python3 scripts/verify_release.py
```

Install the pinned leanprover/lean4:v4.35.0 toolchain. The project has no Mathlib or third-party Lean package. The repository contains plain-language and technical sources, generation scripts, ledgers, theorem references and verification logs. The optional Linux proc-self compatibility shim documented in BUILD_VERIFICATION.md concerns the original execution container; ordinary macOS/Linux and the GitHub workflow use the normal compiler. Document rendering has separate Python/font dependencies.

References

[1] Leo XIV. Magnifica Humanitas. Official English edition, 15 May 2026. Formal scope: §§97-111; contextual discussion: §§10, 14, 47, 128. Accessed 2 November 2026. Official HTML/PDF URLs and source fingerprints: data/source_version.json.

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[4] Travis Darshan. MagnificaHumanitasFormalization, formal checkpoint 0.16, release 1.1: public report 1.5, technical paper 1.2 and one-page summary 1.0. Companion source archive as described above. Repository (access required while private): https://github.com/tdarshan0917-hub/MagnificaHumanitasFormalization.

Use and conclusion

Use the artifact to trace one consequential decision: who is assigned, who can challenge it, which reviewer has scope, how a finding reaches the correcting office, and whether the operative record changes. The rule comparison separately examines criteria against a declared assessment standard. The contribution is the inspectable source/interpretation/proof chain and working comparison models. The practical aim is a protection that reaches the affected person; real accessibility, delivery of support and broader human circumstances require institutional evidence beyond the modeled record update.

Research, formalization and drafting were supported by ChatGPT (OpenAI). The author takes responsibility for interpretation, assumptions and presentation; Lean supplies the stated proof check. These examples support examination of institutional procedures without asserting institutional adoption or endorsement.

16. Connections to human oversight and contestable decisions

This discussion connects the model distinctions to selected European provisions. It is a comparison, not a formalization of those laws. The legal provisions add no premises or theorems to the Lean suite. Whether they apply to a real system depends on their scope, exceptions and applicable legal framework.

Human oversight: EU AI Act, Articles 14 and 26(2)

Article 14 addresses effective human oversight of high-risk AI systems. Article 14(4) includes, as appropriate and proportionate, the ability to interpret outputs, disregard or reverse them, and intervene or stop the system safely. Article 26(2) requires deployers to assign oversight to people with the necessary competence, training, authority and support [5-6]. Our distinction between a board's existence, its decision coverage and its intervention power offers questions for examining implementation. The timely-stop model adds a conditional illustration of when an intervention becomes effective. These results do not establish the legal adequacy of a particular system or require that every oversight arrangement use our example's board.

Contesting a decision: EU GDPR, Article 22

Article 22(1) concerns decisions based solely on automated processing that have legal or similarly significant effects. Paragraph 2 sets exceptions. For the contract-necessity and explicit-consent exceptions in Article 22(2)(a) and (c), paragraph 3 requires safeguards including human intervention, expression of the person's view and contesting the decision. The statutory-authorization exception has its own safeguard requirement in paragraph 2(b) [7]. Our review/appeal distinction helps explain why internal review alone does not show that the person can contest a decision. This is not a universal right to reverse every AI-assisted outcome.

The appeal-to-record-update model then asks a further implementation question: when review warrants correction, does the finding reach an authorized office and change the record? That checked workflow is not itself prescribed by Article 22, and a corrected record does not establish payment or restoration of every harm. The usefulness of the comparison is to identify evidence to examine, while keeping legal applicability and model correctness separate.

Primary sources, checked 3 November 2026

[5] Regulation (EU) 2024/1689, Article 14(1)-(4). European Commission, AI Act Service Desk, article text: https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-14

[6] Regulation (EU) 2024/1689, Article 26(2). European Commission, AI Act Service Desk, article text: https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-26

[7] Regulation (EU) 2016/679, Article 22(1)-(3), EUR-Lex, CELEX 32016R0679: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679.